

Illuminating the dark genome with HERV-aware pangenome graphs

Rigshospitalet MDxCore and HERVolution Therapeutics · EMBO Pangenomics 2026

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Why HERVs?

Human endogenous retroviruses (HERVs) are remnants of ancient retroviral integrations typically fixed in the germline over million of years of primate evolution. Despite comprising **~8% of the human genome**, the linear GRCh38 reference captures only one haplotype and systematically misses polymorphic HERV insertions, including recently active HERV-K (HML-2) elements that remain unfixed in human populations. Combined with structural and evolutionary features (fragmentation, recombination, nesting, and orientation bias), this limits bioinformatic detection and annotation.

HERVs cannot be detected using current linear (GRCh38) or graph-based references (HPRC) and are invisible to standard bioinformatic pipelines.

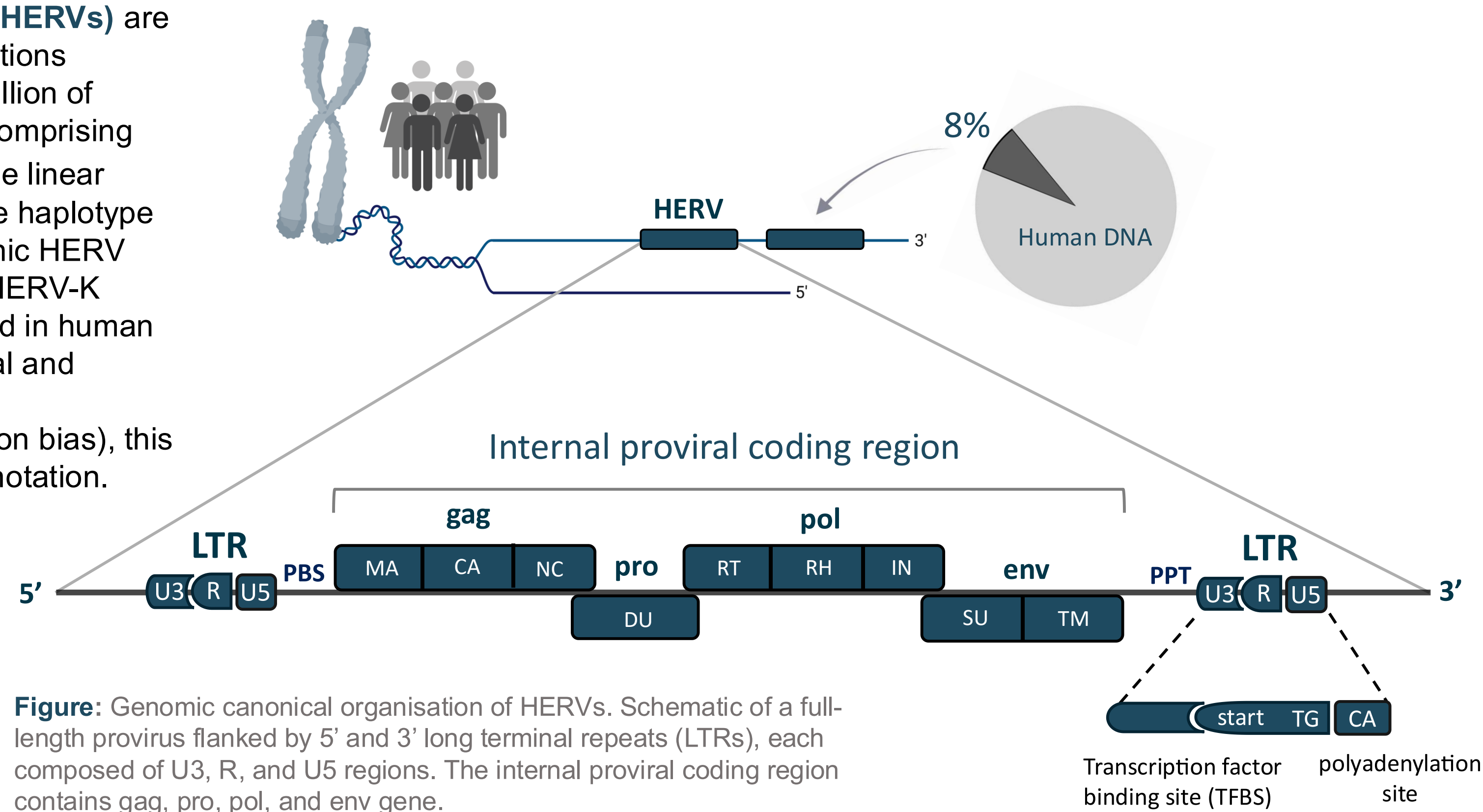
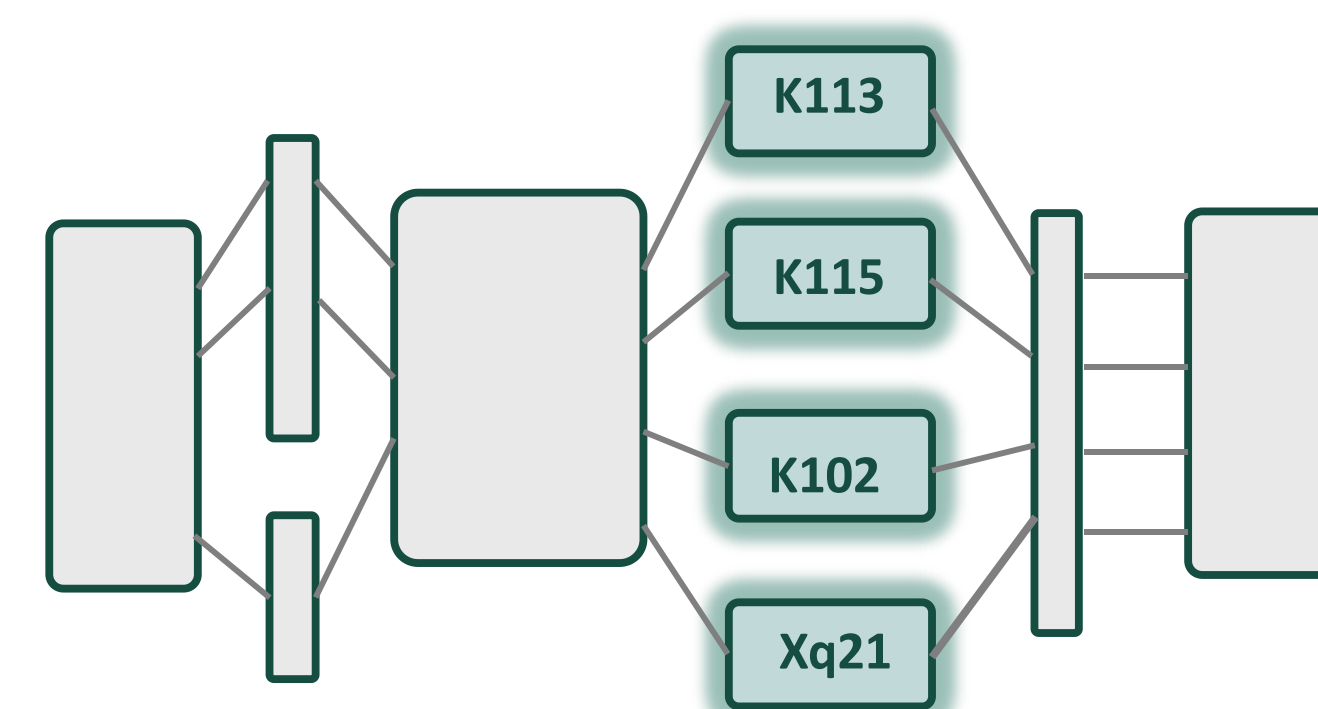


Figure: Genomic canonical organisation of HERVs. Schematic of a full-length provirus flanked by 5' and 3' long terminal repeats (LTRs), each composed of U3, R, and U5 regions. The internal proviral coding region contains gag, pro, pol, and env gene.

Goal

Build a HERV-aware pangenome graph supporting locus-resolved detection and quantification of HERV expression in cancer and metabolic disease



polymorphic HERV insertions

Figure: Alternative proviral paths through shared flanking nodes vs. single linear path in hg38. Each HERV locus should be encoded as alternative paths sharing flanking anchor nodes.

HERV Biology & Disease Relevance

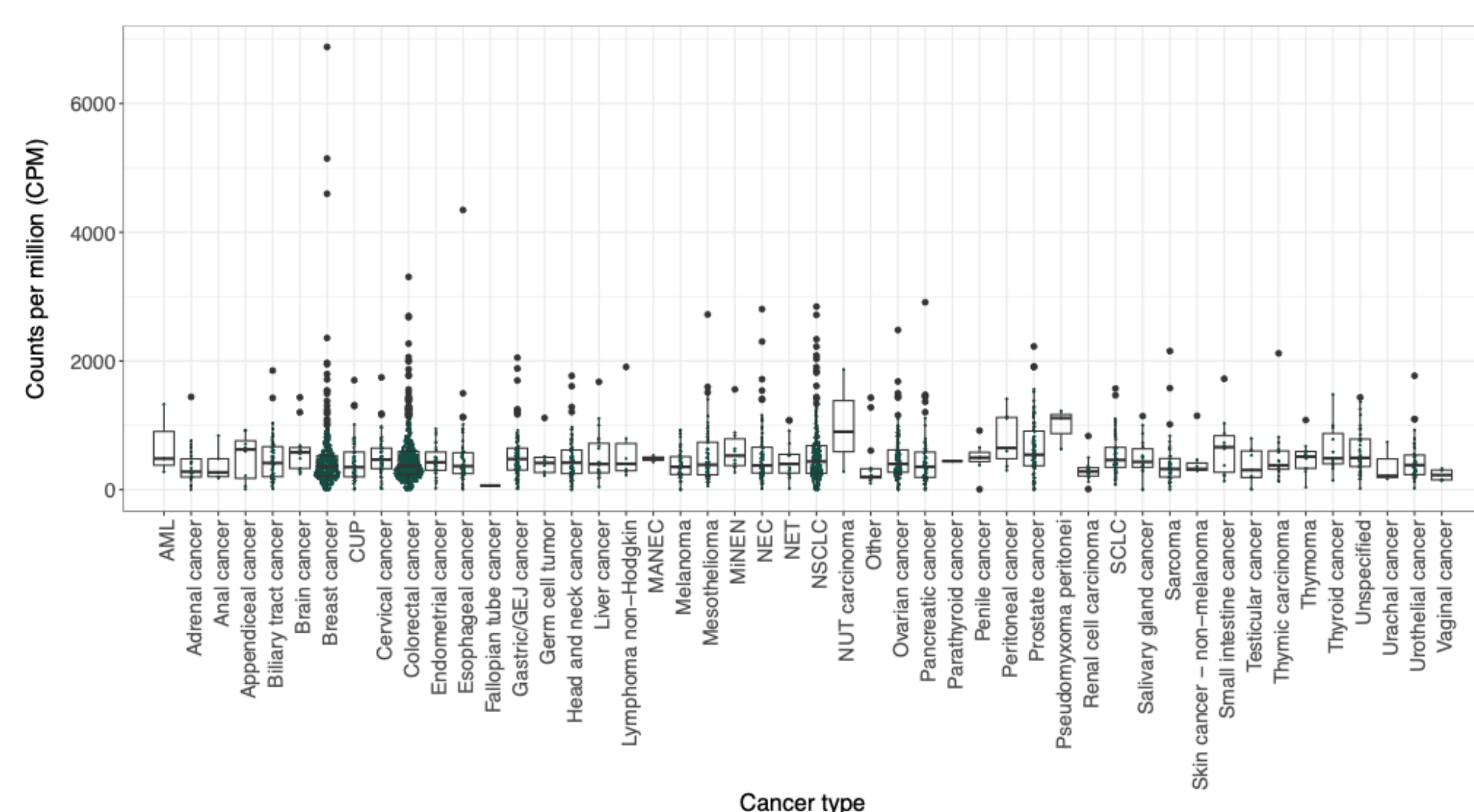


Figure: Transcriptional HERV-K activity across 2685 human tumour biopsies spanning 37 cancer types. RNA-seq was performed at Rigshospitalet as part of the Copenhagen Prospective Personalised Oncology Study (CoPPO) (Belcaid et al. 2025). Median quartiles are illustrated for each cancer type.

Although most remain transcriptionally silent in healthy tissue, **HERV-K (HML-2)** being the most polymorphic and transcriptionally active family, is often expressed and reactivated in cancer, metabolic disease, autoimmunity, and neurodegeneration.

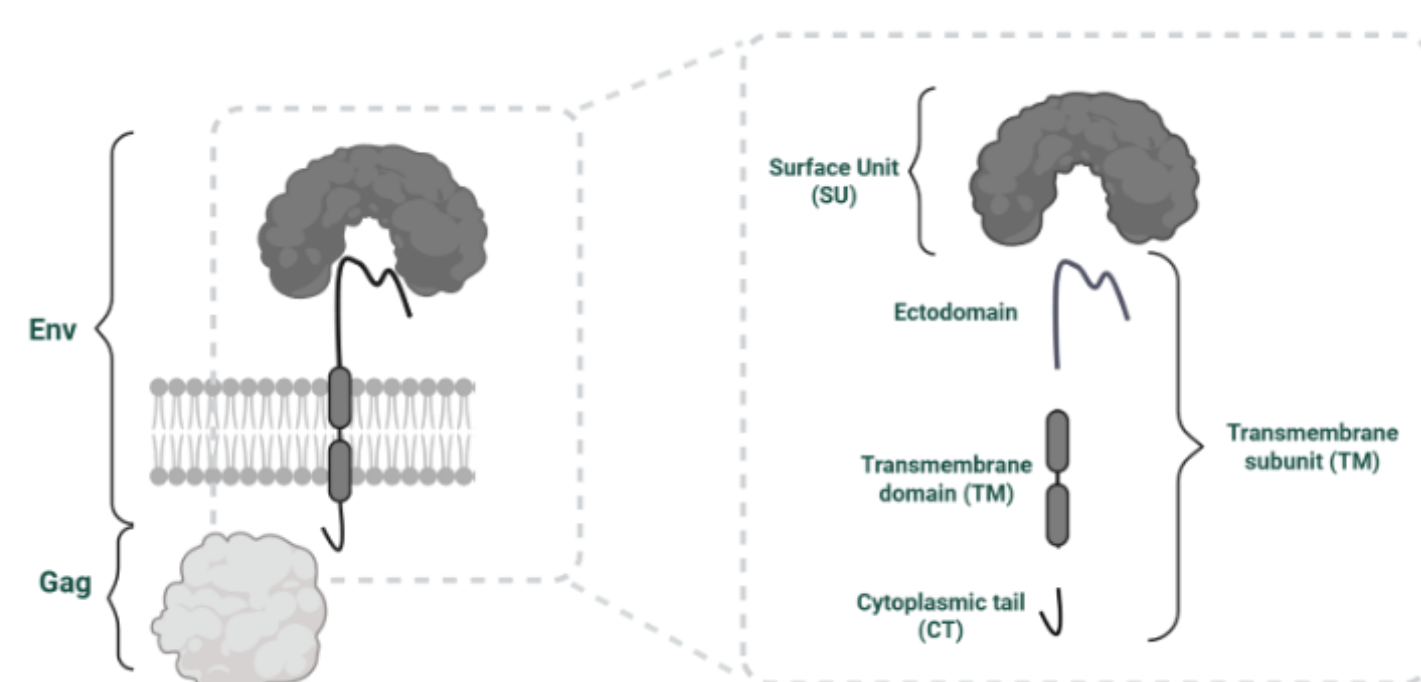


Figure: HERV-K (HML-2) provirus product. Surface unit (SU) mediates receptor binding, the transmembrane subunit (TM) anchors the virus like particle (VLP) in the lipid bilayer.

The Problem

Reference Bias and Multi-mapping

A read originating from a non-reference HERV insertion may be forced onto a paralog that itself has unresolvable multi-mapping:

- Single haploid linear reference cannot represent HERV diversity: insertions absent from GRCh38 are invisible and reads from polymorphic loci misalign to paralogous reference copies.
- 98% identity within the HERV-K (HML-2) family means that short reads cannot be uniquely assigned to a single locus.
- Standard pipelines either (a) discard multimappers, (b) randomly select a primary alignment or (c) report all hits.

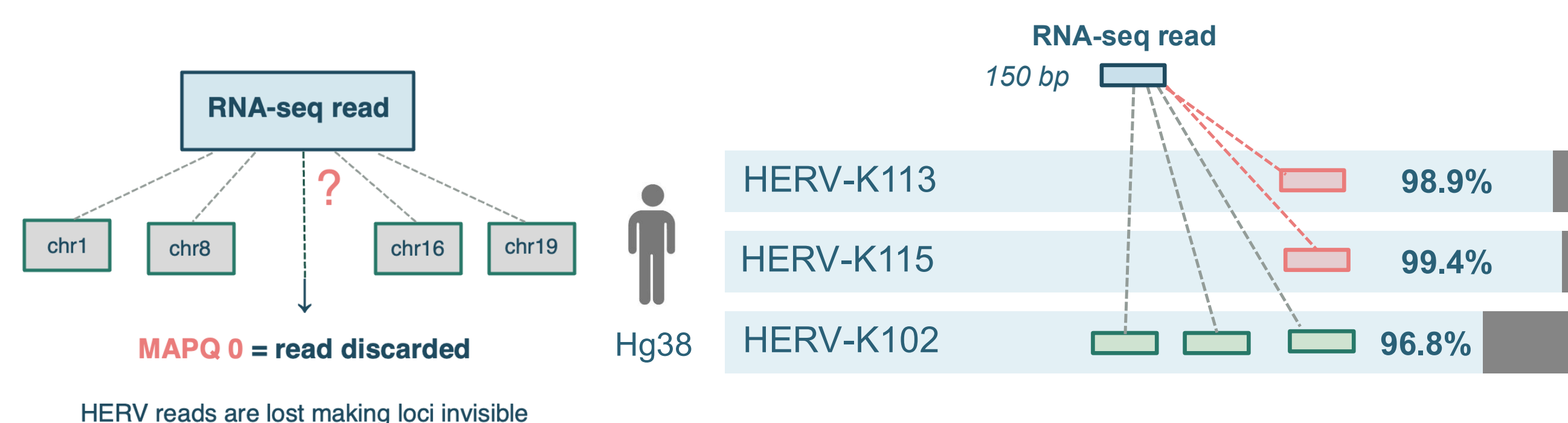


Figure: Fundamental problems for short-read RNA-seq mapping of HERV insertions. Reads are either lost or misaligned since (i) HERV paralogs share above 98% identity across their 9.5 kb length. At 150bp read length many reads match multiple loci equally well (ii) many HERVs are absent from reference genome Hg38.

Pipeline Overview

Step 1: HERV genotyping

HPRCv2 haplotypes are genotyped for HERV-K content to select a diverse subset.

Problem: We want as few haplotypes as possible to keep graph complexity manageable while covering the broad range of HERV insertions and structural variation.

Step 2: Graph Construction

Graph should preserve repetitive and structural elements while maintaining low complexity.

Problem: construction tools handle repetitive elements differently. PGGB collapses them, while minigraph-cactus preserves linearity but treats large insertions as bubbles which can hide intra-family structure.

Step 3: Graph Annotation

Annotation of repetitive variation and regulatory features should i) harmonised labels from established databases and ii) propagate consistently across non-reference insertions on alternative paths.

Problem: Annotation databases like RepeatMasker, Dfam and HERVd disagree on family/subgroup labels and element boundaries.

Step 4: Graph Augmentation

Polymorphic proviruses and non-reference insertions discovered from long-read data should be added as alternative paths in the graph.

Problem: Risk introducing cycles where paralogs share LTRs – there are no established method for propagating annotations from augmented paths back into existing graphs.

Step 5: Locus-Resolved Expression Mapping

The final pangenome graph is to be applied as alternative reference genome for RNA-seq, producing locus-level expression calls.

Problem: Short-read alignment in cyclic graphs is unsolved. vg giraffe and PanGenie require DAG topology, and GraphAligner is build for long-reads.

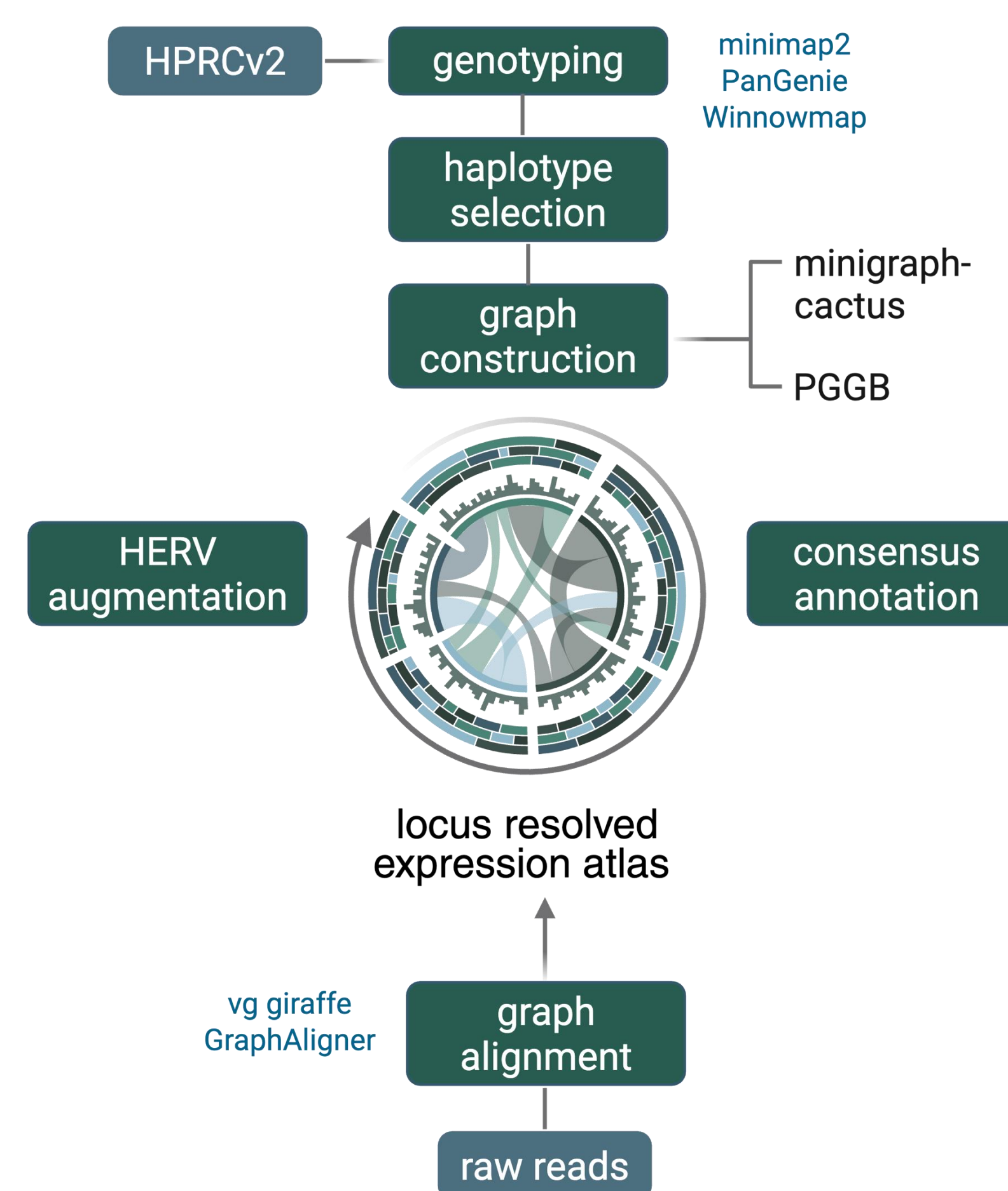


Figure: Pangenome pipeline. HPRCv2 haplotype assemblies are genotyped for HERV-K content and a diverse subset selected for graph construction. The graph is iteratively refined through annotation and augmentation with polymorphic insertions. The final HERV-aware graph serves as a reference for locus-resolved RNA seq alignment.

Open questions and challenges

- How do we balance complexity and structural variation during graph assembly and construction?
- Is there a graph topology that preserves locus identity without inducing cycles?
- Are there parameter regimes specifically tuned for repetitive content?
- Which approaches exist for handling non-reference insertions discovered in alternate haplotypes?
- Long-read index creation with vg giraffe fails and is needed during mapping.
- Why use cyclic alignment (e.g. GraphAligner)?

